VISHWAKARMA AWARDS
A Maker Bhavan Foundation Initiative

MAKER BHAVAN FOUNDATION

VISHWAKARMA AWARDS 2026-27

VK26-IITH-SC-0290

SPECTER

Terrestrial laser scanner for 3D documentation of urban infrastructure

Sustainable Cities · Smart Urban Infrastructure & Asset Tracking

Problem Context: Project Overview

₹25-80 lakh
COMMERCIAL SCANNER COST

₹41,256
SPECTER BILL OF MATERIALS

60-190×
COST REDUCTION

1.2 Million
POINTS, 4-STATION PARK SURVEY



Figure: 250,000-point SPECTER scan of a furnished room, front walls cut away. Desk, monitors, chair and ceiling fan are all resolved.

Laser-scan any building for ₹41,256, not ₹25-80 lakh

- **Title:** SPECTER — a low-cost nodding-LiDAR terrestrial scanner for documenting urban infrastructure.
- **Team ID: VK26-IITH-SC-0290.**
- **Track:** Theme 6, Sustainable Cities — Smart Urban Infrastructure & Asset Tracking.
- **Core innovation:** a 360° direct time-of-flight 2D LiDAR, nodded about an offset axis under drift-free absolute positioning, produces survey-format 3D scans of up to 500,000 points per station.
- **Status:** built and scanning — 250,000-point room scans and a 4-station, 1.2-million-point public park survey, exported in the survey-standard LAS format.
- **Why it matters:** municipal bodies cannot afford laser scanning today; SPECTER makes it a sub - **₹50,000** instrument any engineering office can own.

Built and scanning today: room scans of 250,000 points and a merged 4-station outdoor survey of 1.2 million points — at 60-190× lower cost than a commercial scanner.

Problem Context: Problem Statement & Beneficiaries

No accurate 3D record of civic infrastructure

- **Pain point:** urban local bodies and contractors record drains, culverts, retaining walls and heritage façades with tapes and photos — a sketch with a few dozen dimensions, not complete geometry.
- **Why it persists:** terrestrial laser scanning solves this and is standard practice abroad, but scanners cost ₹25-80 lakh. Adoption is blocked by cost, not by need.
- **Consequences:** retrofits are designed from outdated drawings, condition changes go unmeasured, and after floods or partial collapses there is no baseline to compare against.
- **The gap:** no instrument exists between a tape measure and a ₹25-lakh scanner that a district engineering office can actually buy.

Where SPECTER sits

Manual survey
tape + photos

→

SPECTER
₹41,256

→

Commercial TLS
₹25-80 lakh

Target Beneficiaries

Municipal engineering

As-built records of drains, culverts and civic buildings

Public works departments

Retrofit surveys and condition baselines

Survey & civil contractors

Renovation surveys, stockpile volumes

Heritage conservation

Façade and interior recording in hours, not days

Disaster assessment

Stand-off capture of damaged structures

Polytechnics

An affordable platform for teaching 3D metrology

Key insight: a manual survey records only what someone thought to measure. A scan records everything present, and any dimension can be extracted years later.

User Journey: Before State

30–60 min
MANUAL SURVEY OF ONE ROOM (TYPICAL)

1–2 days
HAND-MEASURED FAÇADE (TYPICAL)

~30
DIMENSIONS CAPTURED PER SURVEY

₹25-80 lakh
BARRIER TO THE SCANNING ALTERNATIVE

Today: a municipal engineer documents a damaged retaining wall

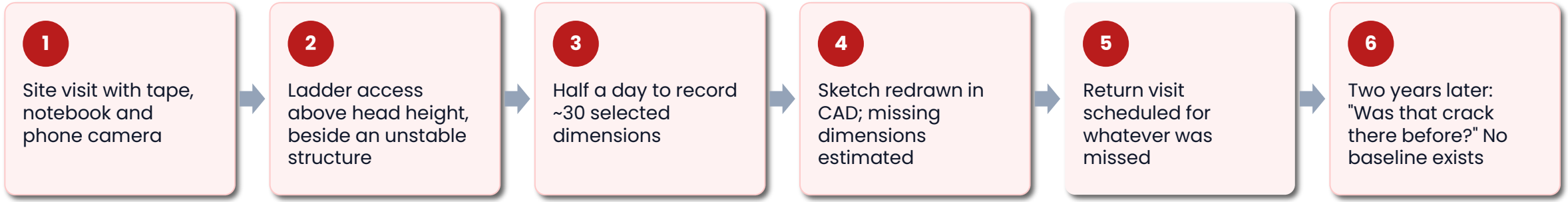


Diagram: AI Generated

- **Time loss:** every survey is slow and every omission costs a repeat site visit.
- **Safety hazard:** measuring above head height means ladder work beside cracked or unstable structures.
- **No baseline:** without complete geometry, movement, settlement or material loss cannot be measured over time.

Current practice: a manual survey captures only the dimensions someone thought to take; anything missed costs another site visit.

User Journey: After State

SPECTER · STAGE 2

81 sec	~2 min	1.2 M	0
125,000-POINT STATION (MEASURED)	250,000-POINT STATION (EST.)	POINTS FROM 4 MERGED STATIONS	LADDERS OR CLIMBS NEEDED

With SPECTER: four tripod stations, nothing climbed

1. Tripod placed; operator levels the head and starts the scan
2. Automatic sweep: 81 s for 125k points, about 2 min for 250k
3. Repeat at 3–4 stations, 5–7 m apart, with overlapping views
4. Stations registered into one cloud; exported as LAS for CAD and GIS
5. Any dimension extracted on demand — no return visit
6. Next year: new scan vs archive; movement shown as a colour map

Diagram: AI Generated

- **Time gain:** a site in minutes per station, with complete geometry rather than selected dimensions.
- **Safety gain:** capture from up to 10 m away on stable ground; nothing climbed, nothing approached.
- **Permanent record:** every scan is a baseline; repeat scans quantify change instead of guessing at it.

Park survey: 4 stations merged



> **Demonstrated, not projected:** four park stations, 5–7 m apart, were scanned and merged into one 1.2-million-point survey with SPECTER.

System Architecture: Block Diagram

POWER

- **3S2P Li-ion pack**
  11.1 V · 6 Ah · BMS cut-off 8.7 V
- **5 A fuse + master switch**
  Peak draw ≈ 2.8 A
- **USB-PD module**
  5 V → Raspberry Pi 5
- **12 V direct rail**
  → TMC2209 motor supply

ELECTRONIC CONTROLLERS

- **Raspberry Pi 5 (8 GB)**
  Capture, transform, export
- **RPLIDAR C1**
  USB · 460,800 baud
- **XIAO ESP32-S3**
  USB CDC · 115,200 baud
- **TMC2209 driver**
  EN / STEP / DIR · 1/8 microstep

ACTUATION & MECHANICAL

- **NEMA 17 stepper**
  1.8° · holding ≥ 4.2 kg·cm
- **GT2 belt, 20T : 60T**
  3:1 · 0.075° per microstep
- **8 mm steel spindle**
  2 × KP08 self-aligning bearings
- **Offset sensor mount**
  50 mm offset + counterweight

FIRMWARE & LOGIC

- **XIAO firmware**
  Absolute moves · ramps · limits
- **C1 protocol driver**
  Header check · byte resync
- **Step-settle capture**
  Angle-tagged revolutions
- **Transform + export**
  PLY · LAS/LAZ at 1 mm

Capture loop

Command angle → Wait for ARRIVED → Discard partial rev → Record full revs → Transform to XYZ → Next angle

Diagram: AI Generated

Architecture rationale: Linux can pause any process for milliseconds and silently drop stepper pulses, so step generation runs on a dedicated microcontroller; the Pi tags each revolution with the angle the controller confirms.

System Architecture: Specifications & Hardware

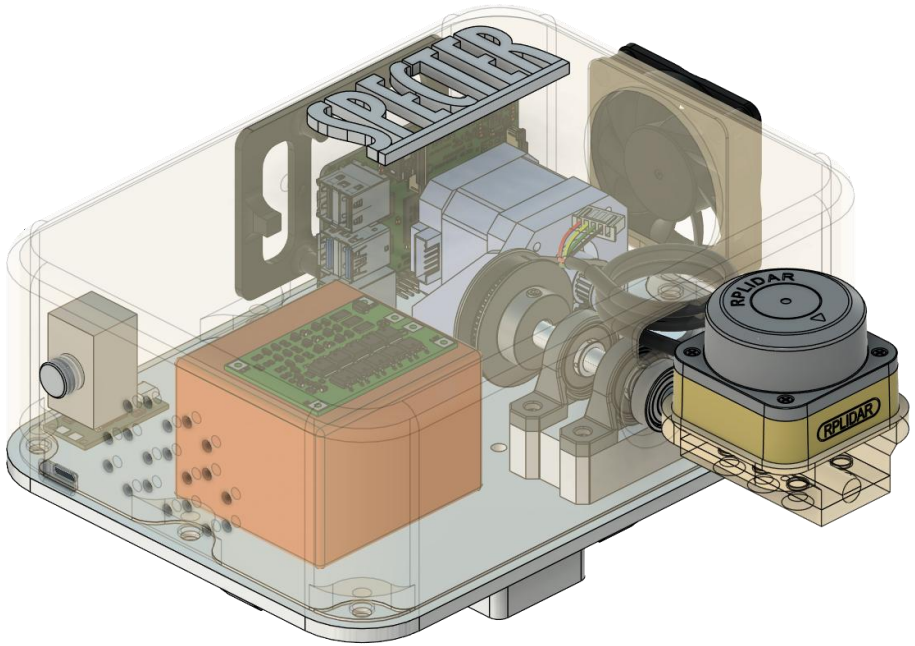
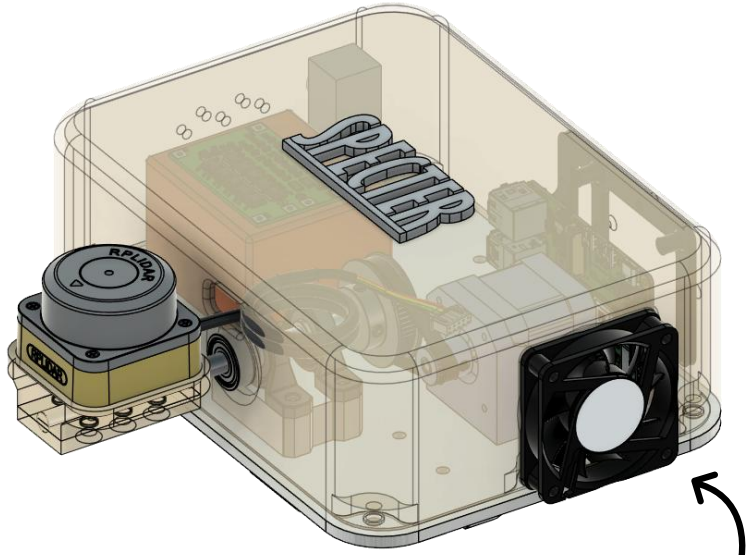


Figure: SPECTER in Fusion 360 — RPLIDAR C1 on the offset spindle, Pi 5, drive, 3S pack and cooling fan outside the enclosure.

Parameter	Value	Selected hardware / protocol
Range	0.05–12 m (spec)	RPLIDAR C1, DTOF, 360°
Range accuracy	±30 mm (spec); validation underway	RPLIDAR C1
Valid returns	461 per revolution (measured)	Custom driver, 460,800 baud
Points per station	125k / 250k (measured) · 500k (est.)	1–2 revolutions per position
Multi-station survey	4 stations → 1.2 M points (measured)	Point-pair + ICP registration
Nod resolution	0.075° per microstep	NEMA 17 + 3:1 GT2 + TMC2209
Angular error	±0.11°, non-accumulating (measured)	Absolute positioning, XIAO ESP32-S3
Nod range	±135° soft limits	Firmware-enforced
Compute supply	4.81 V, no throttling (measured)	Raspberry Pi 5, 59.3 °C peak
Energy	66.6 Wh · ~2.5 h (est.)	3S2P 18650, BMS cut-off 8.7 V
Output	PLY + LAS/LAZ, 1 mm precision	plyfile, laspy
Daylight	40,000 lux (spec)	Direct time-of-flight ranging

Every value is labelled: measured on our own hardware, taken from the manufacturer's specification, or marked as an estimate.

CAD Design: Renders & Material Selection



Exploded view

Enclosed assembly

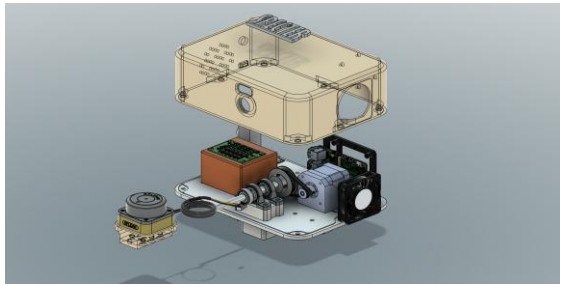


Figure: Fusion 360 assembly — enclosure, spindle in KP08 bearings, belt drive and electronics.

Offset roll-axis spindle: zero self-occlusion by design

Part	Material	Why
Spindle shaft	8 mm precision-ground steel	Stiff (E = 200 GPa); the overhung sensor barely deflects
Bearings	KP08 zinc-alloy pillow blocks	Self-aligning race absorbs misalignment of printed mounts
Drive	GT2 6 mm belt, 20T : 60T	Negligible backlash; tolerant of centre-distance error
Structure	PLA (prototype)	Fast, accurate iteration; PETG / ASA for outdoor builds
Enclosure	Printed shell + cooling fan	Protects electronics; forced air for Pi 5 and driver
Damping	TPU	Absorbs motor vibration; grips the tripod head
Fasteners	M3 + brass heat-set inserts; M5 bolts	Repeated assembly without stripping printed threads

Design iteration: our first concept used a yoke-mounted pitch axis whose arms sat inside the scan volume. The roll-axis spindle offsets the sensor 50 mm, so every structural part sits below the scan plane.

CAD Design: Load & Stress Analysis

8 mm steel shaft in two KP08 bearings; sensor overhung at the tip

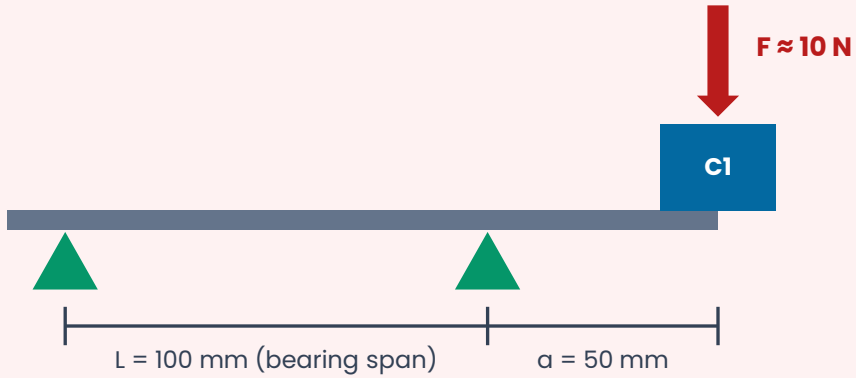


Diagram: AI Generated

Hand-calculated load case

Moving mass ≈ 0.5 kg (est.) × 2 dynamic factor → F ≈ 10 N, at a 50 mm overhang beyond a 100 mm bearing span. Steel shaft, d = 8 mm, E = 200 GPa, $I = \pi d^4/64 = 201\ mm^4$.

Check	Result	Limit / margin
Shaft bending stress $\sigma = F{\cdot}a{\cdot}(d/2) / I$	9.9 MPa	Steel yield ≥ 350 MPa → FoS > 35
Tip deflection $\delta = F{\cdot}a^2(L + a) / 3EI$	0.031 mm	Far below ±30 mm sensor noise
Tip slope (pointing) $\theta = F{\cdot}a(2L + 3a) / 6EI$	0.042°	17× finer than 0.72° scan resolution
Drive torque NEMA 17 holding × 3:1	1.24 N·m	Worst imbalance 0.15 N·m → 8× margin

Mechanism does not limit accuracy: pointing error from shaft bending (0.042°) is 17 times smaller than the sensor's own 0.72° angular resolution.

Electronics: Wiring, Pinout & Power Distribution

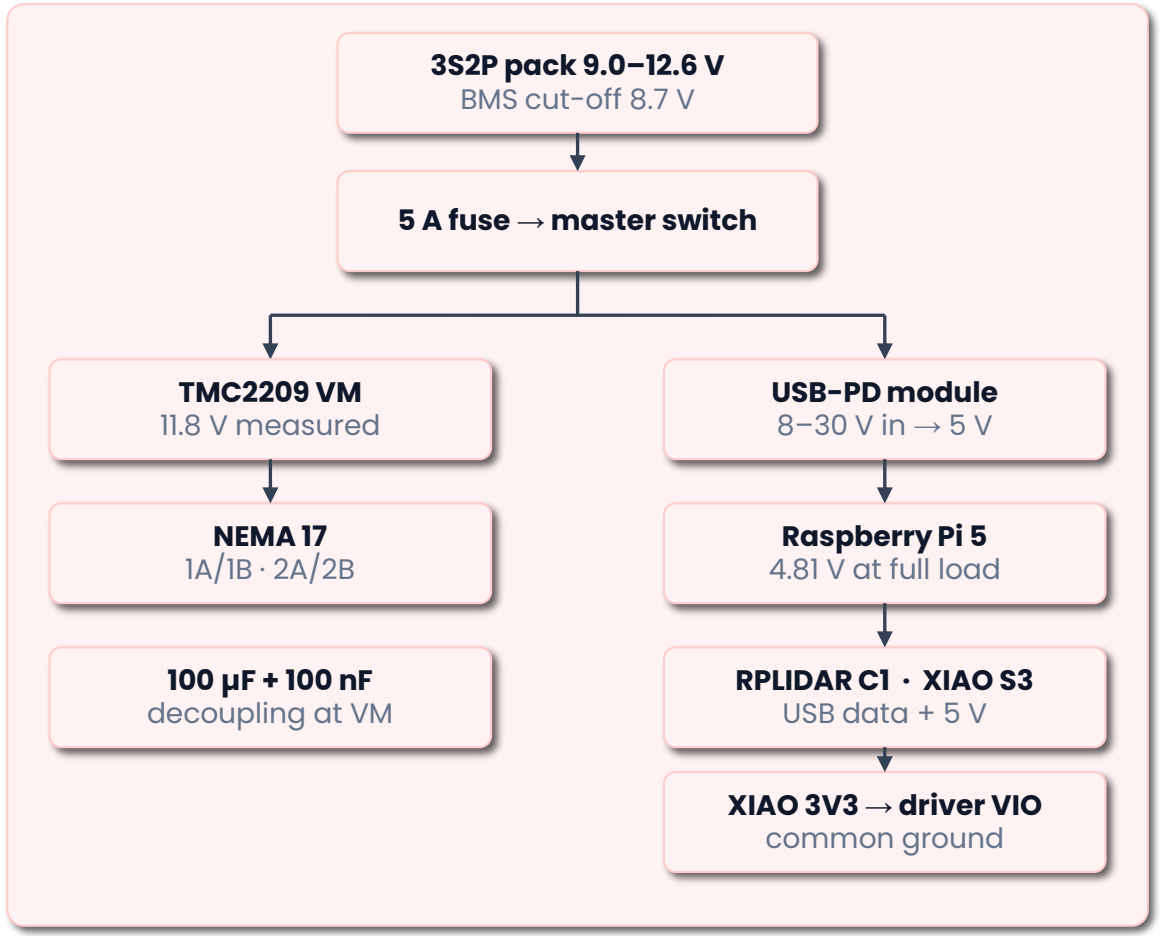


Diagram: AI Generated

Pinout & Signal Map

Signal	From	To	Note
EN	XIAO D0	TMC2209 EN	Low = enabled
STEP	XIAO D1	TMC2209 STEP	Trapezoidal ramp
DIR	XIAO D2	TMC2209 DIR	Positive = clockwise (rear view)
VIO / GND	XIAO 3V3 / GND	Driver VIO / GND	Shared with pack -
VM	11.1 V rail	TMC2209 VM	100 µF + 100 nF
Coils	Driver 1A/1B, 2A/2B	NEMA 17	Pairs by resistance, 2–5 Ω
Lidar data	RPLIDAR C1	Pi 5 USB	CP2102N, 460,800 baud
Control	XIAO ESP32-S3	Pi 5 USB	USB CDC, 115,200 baud

Driver configuration: 1/8 microstepping (MS1/MS2 floating), V_ref set to 0.64 V.

Custom PCB / DRC: Not Applicable — commercial modules wired point-to-point, so there is no PCB layout to rule-check.

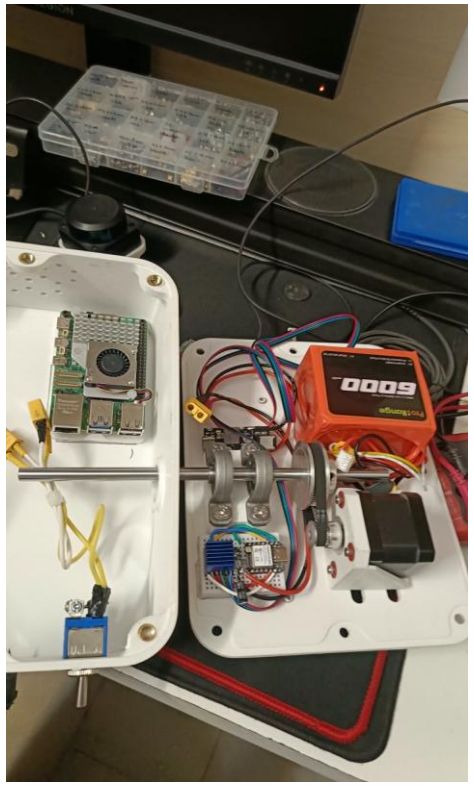
Measured on the integrated system: motor rail 11.8 V; Pi 5 rail 4.81 V under full CPU load with no undervoltage events; peak temperature 59.3 °C.

Electronics: Itemised Bill of Materials

Component	Specification	Qty	Source	Cost (₹)
RPLIDAR C1	DTOF 2D LiDAR, 12 m range, 5 kHz sampling	1	Tanna TechBiz (India)	7,300
Raspberry Pi 5 (8GB)	Quad-core Arm Cortex-A76 @ 2.4 GHz, 8 GB RAM; runs capture, transform and export	1	Robu.in (India)	20,000
Seeed Studio XIAO ESP32-S3	Microcontroller: real-time stepper control (EN / STEP / DIR)	1	Robocraze.com	1,000
SD Card (64 GB)	microSD card (64 GB): OS and scan-data storage	1	Amazon.in	1,600
Tripod	Tripod for Lidar Mounting	1	Local Source	4,500
Stepper motor	NEMA 17, 1.8°, ≥ 4.2 kg·cm	1	Robu.in (India)	749
Motor driver	TMC2209 V2.1 module	1	Robu.in (India)	870
Timing pulleys	GT2 20T 5 mm bore + 60T 8 mm bore	1 + 1	Robu.in (India)	234
Timing belts	GT2 6 mm closed loop, assorted lengths	3	Robu.in (India)	141
Shaft	8 mm precision ground steel	1	Robu.in (India)	279
Bearings	KP08 pillow block, 8 mm bore	2	Robu.in (India)	128
Battery	3S2P 18650 Li-ion, 11.1 V, integrated BMS	1	Robu.in (India)	1,679
Charger	USB-C to 3S 12.6 V boost charger	1	Robu.in (India)	249
Pi power	USB-PD 65 W module (XPM52C, PD 3.1)	1	Robu.in (India)	249
Protection	5 A fuse + holder	1	Robu.in (India)	60
Cooling	Raspberry Pi 5 official active cooler + Cooling Fan	1 +1	Robu.in (India)	718
Miscellaneous Items	Push-button switch, wiring, connectors, fasteners, heat-set inserts, capacitors, filament	-	Robu.in (India)	1,500
Total				**₹41,256**

Procurement status: every component is sourced domestically through various Indian vendors. The design depends on no imported or custom-fabricated parts.

Maker Journey: Authenticity Evidence



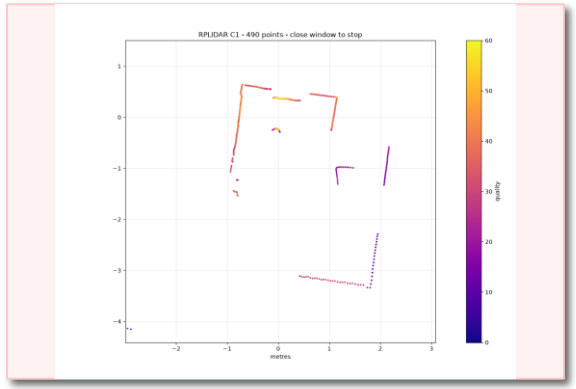




Figure: Build evidence: assembled electronics, first RPLIDAR C1 bench scan on our hardware (490 points, coloured by return intensity), procured 6,000 mAh pack.

Built, wired, coded and scanned by our team

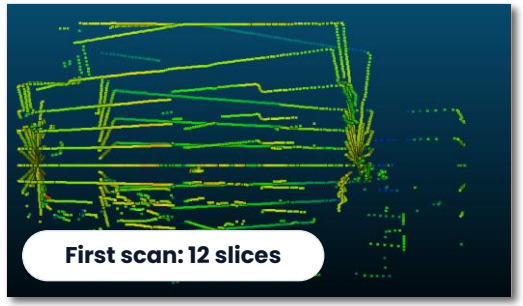
- **Repository:** GitHub— firmware, capture software, CAD (.f3d), BOM, sample scans (.bin).
- **Demo video:** Demo — the head sweeping
- **Fabrication:** photos of team members printing, wiring and assembling, with ID badges.
- **Design files:** Fusion 360 timeline and feature tree of the full assembly.
- **Software:** live scan log from the Pi, with saved PLY and LAZ output.
- **EDA / DRC:** Not Applicable — no custom PCB.

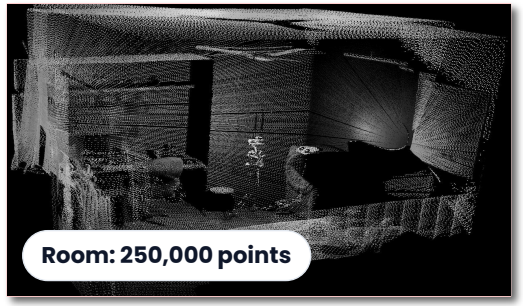
Links

- **GitHub –** https://github.com/HellbeE908/SPECTER
- **Demo Video –** https://drive.google.com/file/d/1721Sx0ei6ZV6otYKVASYwu00BhRb3EuO/view?usp=sharing
- **Drive –** https://drive.google.com/drive/folders/1PsYOReHO6t3g_3KUsVLf8GWxNv96WUw?usp=sharing

Audit trail: files in repository and Drive document build progress across Stage 2.

Maker Journey: Stage 1 → Stage 2 Iterations







Seven decisions changed by testing

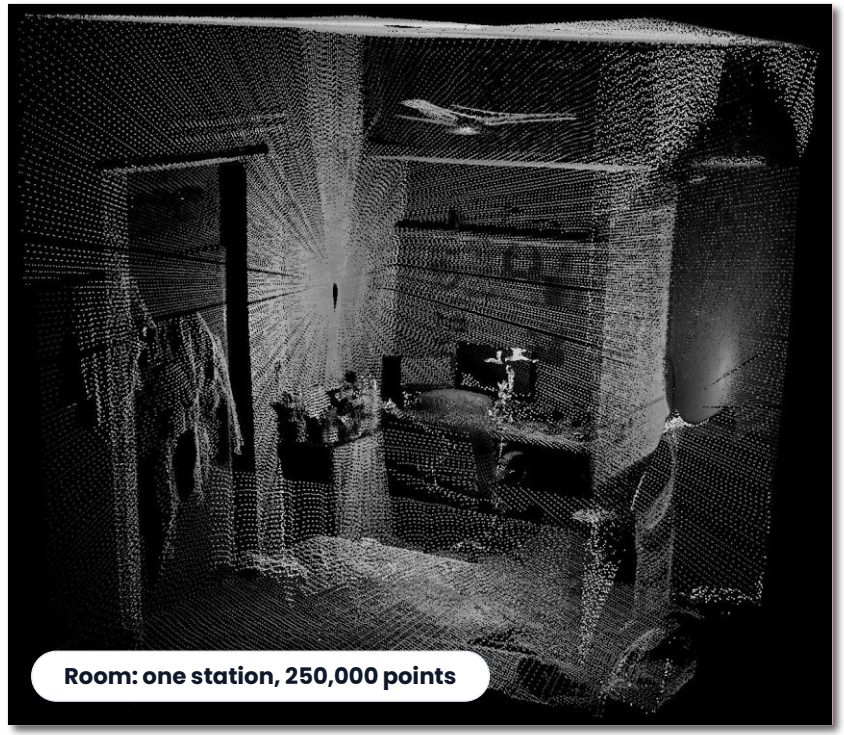
- **Silent drift eliminated:** incremental stepping drifted 3.6° per revolution while reporting correct angles. Absolute-target positioning bounds error at ±0.11°, with no accumulation.
- **Own sensor driver:** a popular Python lidar library failed silently on the Pi. Byte-level tests showed the C1 needs an explicit motor-start command, so we wrote our own protocol driver.
- **Camera removed:** the return signal carries 47–57 distinct intensity levels — enough to render solid-looking surfaces with Eye-Dome Lighting and no camera.
- **Mirror-image bug caught:** a bearing test showed the C1 measures clockwise; without the fix every scan would have been a perfect mirror image.
- **Offset calibrated from data:** a seam across the walls disappeared at a 50 mm offset, measured from the scan itself rather than the CAD.
- **Driver protection:** two drivers were destroyed by inductive spikes when motor wiring was disturbed under power. A strict power-down rule and local decoupling followed; no failures since.
- **Compact controller:** moved from an Arduino Nano ESP32 to a XIAO ESP32-S3 to fit the compact base.

Verified results: 461 valid returns per revolution · 250,000-point stations · 4-station outdoor merge of 1.2 M points · angular error held at ±0.11° · no undervoltage at full load.

Results: What SPECTER Captures Today



Park: 4 stations merged, 1.2 M points



Room: one station, 250,000 points

1.2 M
POINTS · 4-STATION PARK MERGE

250,000
POINTS · SINGLE ROOM STATION

0
CAMERAS — LASER INTENSITY ONLY

LAS / LAZ
SURVEY-STANDARD OUTPUT, 1 MM precision

Real captures only: benches, slats and tree trunks stay single and crisp across all four merged stations — the alignment holds. Every image is a SPECTER scan, not a photograph.

Applications Beyond the Urban Track

SPECTER · STAGE 2

Already demonstrated outdoors



- **Same instrument, same pipeline:** a 4-station park survey (1.2 M points) already resolves individual trunks, benches and canopy.
- **No new hardware:** every site here uses the same scanner, battery and LAS output.
- **Rated honestly:** each application shows where SPECTER fits and where it stops.

Caves & tunnels

STRONG FIT

- The laser brings its own light: works in total darkness
- No GPS needed; battery-powered; most passages lie within 12 m

Limit: Humidity and dust need the enclosure; long passages need many chained stations

Archaeology

GOOD FIT

- Non-contact: nothing fragile is touched
- Trenches, interiors and site layout; repeat scans track excavation

Limit: ±3 cm captures structure, not fine carvings or inscriptions

Forestry & parks

DEMONSTRATED

- Park survey already resolves individual trunks
- Tree counts, trunk diameter and canopy extent

Limit: Wind blurs foliage; trunks measure reliably, canopy less so

Natural terrain

PARTIAL FIT

- Change detection on landslide scarps, riverbanks and quarry faces
- Repeat surveys measure erosion over time

Limit: 12 m range limits area; sun and vegetation cut returns — the 30 m S2 lidar extends reach

Focus stays on urban infrastructure: these sites reuse the same instrument and pipeline, each rated by fit with its limits stated.

Safety & Hazard Mitigation

Design FMEA (Failure Mode and Effects Analysis)

Failure mode	Hazard	Mitigation
Missed motor steps	Silently distorted scan	Absolute positioning, acceleration ramps, settle delay; return-to-mark test before scanning
Motor wiring disturbed under power	Driver destroyed by inductive spike (occurred twice)	Power-down rule for all motor wiring; 100 µF + 100 nF decoupling at the driver
Pi supply sag	Throttling, lidar dropouts	USB-PD supply + current override; undervoltage flag checked under load (0x0 recorded)
Lidar data corruption	Misplaced points	Header check, byte-level resynchronisation, timeout with partial save
Battery over-discharge	Cell damage, fire risk	BMS cut-off at 8.7 V; scans stopped above cut-off
Power cut during file write	Corrupted SD card	Shutdown procedure before switch-off

Physical fail-safes

- 5 A fuse on the pack positive (peak draw ≈ 2.8 A); single master switch
- BMS: over-discharge cut-off 8.7 V, over-current and short-circuit protection
- Firmware soft limits ±135° reject out-of-range moves, protecting the cable loop
- Ctrl+C abort returns the head to level and saves captured data
- Emergency stop: unplugging the controller USB halts motion instantly

Standards considered

- **Laser:** Class 1 eye-safe C1 (IEC 60825-1, manufacturer rating) — no exclusion zone needed
- **Data:** ASPRS LAS 1.2 output at 1 mm precision, readable by survey and GIS tools
- **Power:** USB Power Delivery 3.1 certified controller in the Pi supply
- **Ingress:** not yet rated; fan-cooled enclosure designed, sealed variant for outdoor use planned

Safety by design: stand-off measurement is itself the main safety benefit — no ladder work beside damaged or unstable structures.

Roadmap: Plan Against the Competition Timeline



Diagram: AI Generated

Oct – Nov 2026

Prototype validated

- Done: 250k-point room scans; 4-station park merge, 1.2 M points; LAS export
- Next: wall-flatness RMS, tape-checked dimensions, bidirectional repeatability

DELIVERABLE
Published accuracy figures

Dec 2026

Automation & change detection

- Automatic multi-station registration (Open3D)
- Displacement tests at 20, 50, 100, 200 mm
- Print and fit the designed enclosure

DELIVERABLE
Detection threshold in mm

Jan 2027

Standalone field kit

- Phone control over the Pi's own hotspot
- On-device registration and LAS export
- IMU auto-levelling

DELIVERABLE
Laptop-free operation

Feb 2027

Field validation & finale

- Real structure vs tape / total station
- Noon vs dusk daylight test
- Live demo: scan, move an object, show the change

DELIVERABLE
Grand-finale live demo

Grand-finale commitment: a battery-powered, phone-operated scanner with published accuracy — a site scanned live, an object moved, and the change measured on screen.

Technical Challenges & Risks



Figure: one room station. Pole crowding on the right wall, and the occlusion shadow behind the bed, where the scanner cannot see.

Resolved during Stage 2 (observed)

- **Silent angular drift:** 3.6° per revolution, invisible to the controller.
- **Sensor would not spin on Linux:** a standard library read device health, yet every scan request failed.
- **Driver destruction:** two TMC2209s lost to inductive spikes; one also damaged a controller supply rail.
- **Moving objects:** a running ceiling fan appeared as ghost fragments across slices.

Open risks for field use

- **Pole crowding & occlusion:** one station leaves dense poles and shadows; merging fixes this but is manual today.
- **Sunlight:** rated 40,000 lux; midday sun in India can exceed twice that.
- **Heat:** PLA softens near 60 °C, reachable in direct summer sun.
- **Manual registration:** point-pair picking per station takes skill and time.
- **Manual levelling:** ±1° phone inclinometer ≈ 9 cm tilt over 5 m.

Found on the bench, not in the field: testing exposed four failure modes before any field deployment, each now fixed and verified.

Potential Solutions & Risk Mitigation

Bottleneck	Mitigation	Status
Silent angular drift	Each move computed from the absolute target angle; error bounded at half a microstep	**Verified · ±0.11°, no accumulation**
Sensor would not spin	Custom protocol driver sends an explicit motor-start command before scanning	**Verified · 461 returns / rev**
Driver destruction	Power-down rule for motor wiring; 100 µF + 100 nF decoupling; spare driver held	**Verified · no failures since**
Moving objects	Scan protocol: fans off, doors fixed, operator out of view	**Verified · fan sharp at 250k**
Pole crowding & occlusion	Overlapping stations 5–7 m apart fill shadows and even out density	**Verified · 4-station park merge**
Manual registration	Automatic pipeline: FPFH + RANSAC coarse alignment, then ICP (Open3D)	**Planned · Dec 2026**
Sunlight saturation	Morning, evening or overcast scanning; noon-vs-dusk comparison test	**Planned · Feb 2027**
PLA heat softening	Outdoor parts and enclosure printed in PETG (~80 °C) or ASA (~100 °C)	**Planned · Dec 2026**
Manual levelling error	Full 3-axis ICP corrects residual tilt; IMU auto-levelling	**ICP in use · IMU Jan 2027**

Fallback options: a longer-range DTOF lidar (SLAMTEC S2, 30 m) drops into the same mechanism with one driver change; natural features (trunks, benches) or portable box targets anchor registration on open sites.

Empirical verification: before the fix, every 10° command moved only 9.9° (3.6° short per turn); after it, logged error stayed within ±0.11° and read exactly zero at 270°.

References & Sources

Research

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Datasheets & standards: SLAMTEC RPLIDAR C1 datasheet · Analog Devices (Trinamic) TMC2209 datasheet · Raspberry Pi 5 documentation · ASPRS LAS Specification 1.2

Image sources: all photographs, CAD renders and point-cloud scans are the team’s own. Diagrams marked “AI Generated” were drafted with AI assistance.